Calcurion Physics Worksheets

A-Level Physics

Topic: Thermal Physics

MARKSCHEME

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1. Sum of random KE and PE of molecules **A2** [2]
2. $\Delta U = Q + W$ (or $Q - W$ depending on sign convention) **A2** [2]
3. Obeys gas laws perfectly; no intermolecular forces; point particles **A2** [2]
4. $pV = nRT$ **A1** [1]
5. $p = 9.97 \times 10^4$ Pa **M1A1** [3]
6. $p \propto 1/V$ at constant T **A1** [1]
7. 400 kPa **M1A1** [2]
8. Energy to raise 1 kg by 1 K **A1** [1]
9. 1.26×10^5 J **M1A1** [3]
10. Energy to change 1 kg solid to liquid at constant T **A1** [1]
11. 6.68×10^4 J **M1A1** [2]
12. $\langle KE \rangle \propto T$ **A2** [2]
13. $c_{\text{rms}} = \sqrt{3RT/M} = 483$ m/s **M1A1** [3]
14. Random motion of particles; evidence for molecules **A2** [2]
15. Hyperbola; $pV =$ constant **A2** [2]
16. Momentum change on walls; pressure from force rate **A4** [4]
17. From $pV = \frac{1}{3}Nm\langle c^2 \rangle$ and $pV = NkT$ **A3** [3]
18. $W = nRT\ln(V_2/V_1) = 3450$ J **M1A1** [4]
19. Temperature falls; work done by gas from internal energy **A3** [3]
20. $W = 0$; $Q = \Delta U$ **A3** [3]
21. $\Delta U = 300$ J **A2** [2]
22. At constant p, gas expands and does work; extra energy needed **A3** [3]
23. 9600 J **M1A1** [2]
24. Heater, thermometer, insulation; $mc\Delta\theta = IVt$ **A4** [4]
25. Theoretical temperature at which molecular KE is zero **A2** [2]